

defence_script_15min

Thesis defence — 15-minute speaking script

Luca Manzi · English · presentation_4 · 20 main slides

Read only the **Spoken script** sections aloud. Bracketed directions are silent cues. **Details to keep in mind** are preparation for questions, not additional material for the timed talk. The spoken text is approximately 1,600 words. Timings include pointing, slide changes and short pauses; rehearse aloud to calibrate them to your pace. Do not read the equations in the technical notes symbol by symbol.

This script follows the current main deck. The detailed sections after each spoken passage are preparation material and are not intended to be delivered verbatim.

The central thread is: **a sensor must retain useful sensitivity through preparation, restricted access and measurement, over the range where it will operate.**

Timing map

Slide	Topic	Duration	Clock
1	Opening	0:30	0:00–0:30
2	Two applications	0:30	0:30–1:00
3	Classical Fisher information	0:50	1:00–1:50
4	Measurement animation and QFI	1:00	1:50–2:50
5	Mixed states and spectral estimation	1:05	2:50–3:55
6	Numerical and circuit pipeline	1:05	3:55–5:00
7	Magnetometry divider	0:05	5:00–5:05
8	Magnetometer model	0:55	5:05–6:00
9	Broader thermal sensitivity	1:00	6:00–7:00
10	Stronger thermal peak	0:55	7:00–7:55
11	Exoplanet divider	0:05	7:55–8:00
12	Imaging problem and SPADE	0:50	8:00–8:50
13	Mode probabilities and cross-talk	0:45	8:50–9:35
14	Available data and the original TQFI goal	0:45	9:35–10:20
15	Why populations are not enough for TQFI	0:40	10:20–11:00
16	From probabilities to a quantum description	0:40	11:00–11:40
17	Physical source state	0:40	11:40–12:20
18	The receiver as a channel sequence	0:50	12:20–13:10
19	Mean-response result and implications	1:00	13:10–14:10
20	Conclusions	0:50	14:10–15:00

Slide 1 — Opening

Spoken script

Good morning. My thesis studies how to assess the sensitivity of quantum sensors when we include the conditions under which they actually operate.

The work was developed in collaboration with the Italian Space Agency. I will connect two applications: a quantum magnetometer and an optical experiment motivated by exoplanet detection.

The common question is: how much information about an unknown parameter can we actually access?

Details to keep in mind

Intuition — preparation only, not extra spoken text: The two applications ask the same question at different stages: what sensitivity does the model allow, and how much can the actual measurement reveal?

- The magnetometer is a numerical model and a circuit-method benchmark. The optical application uses a completed laboratory experiment motivated by exoplanet imaging; it is not an astronomical detection of a real exoplanet.
- Your contribution combines implementations, numerical studies, data interpretation and a future proposal. These have different evidential status throughout the talk.

Slide 2 — Applications

Spoken script

I study this question in two complementary settings.

The magnetometer provides a controlled model: I can vary preparation, noise and access, and compare the resulting sensitivities.

The optical experiment asks a different question: given a real measurement record, which parts of the sensing process can we reconstruct or meaningfully model?

Both start from the same distinction between information in a state and information in recorded outcomes.

Details to keep in mind

Intuition — preparation only, not extra spoken text: The magnetometer asks “what happens when I change the design?” The optical record asks “what can these particular measurements tell me about the design?”

The two applications do not both demonstrate experimental TQFI. The numerical study benchmarks metrological quantities; the archived optical counts do not reconstruct an optical state or its TQFI. Presenting this asymmetry explicitly strengthens the scientific story.

Slide 3 — Classical Fisher information

Spoken script

To connect these applications, consider an ordinary measurement.

[Point to the probability distributions.]

An unknown parameter, such as a magnetic field, determines the distribution of possible outcomes. If changing the parameter barely changes that distribution, the measurement tells us very little. If a small change produces a distinguishable response, estimation becomes easier.

Fisher information quantifies this local sensitivity. It compares the change in the outcome probabilities with their statistical fluctuations.

So it is not simply a measure of how large the signal is. A bright but almost constant signal can be less informative than a weaker signal that responds strongly to the parameter.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Imagine two slightly different field values. If their outcome distributions overlap almost completely, many repetitions are needed to tell them apart. Fisher information quantifies local distinguishability, not brightness alone.

For a fixed measurement with probabilities $(p(x|\theta))$,

$$F_C(\theta) = \sum_x \frac{[\partial_\theta p(x|\theta)]^2}{p(x|\theta)}.$$

The classical Cramér–Rao bound is $\text{Var}(\hat{\theta}) \geq 1/[NF_C(\theta)]$ for N independent repetitions and an appropriate locally unbiased estimator, under regularity assumptions. Large Fisher information is a precision benchmark, not a guarantee that an arbitrary finite-sample estimator attains the bound. “Local” means around a parameter value; it does not establish global identifiability.

Source: [MathBackground.tex](#).

Slide 4 — Fisher information and measurements

Spoken script

A quantum state can contain information that a particular measurement fails to reveal.

[Start at zero degrees. Point to the two coloured vectors.]

These vectors illustrate states at two parameter values. Keep the states fixed, and rotate only the measurement basis.

[Advance through thirty and forty-five degrees.]

The classical Fisher information changes because the measurement probes the state differently.

[Advance through sixty and seventy-five degrees.]

Some bases are more responsive to the parameter variation than others.

[Advance to ninety degrees. Pause at the meeting of the curves.]

Here, the measurement reaches the quantum Fisher information, or QFI: the best local sensitivity available from the state over all measurements.

The state has not become more informative. We have become better at reading the information it already contains. That distinction will matter again in the optical experiment.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Think of looking at a small movement from different directions: one view may hide a change that another reveals. The quantum example concerns changes in outcome probabilities, not literally a two-dimensional geometric shadow.

The current animation uses **mixed states**, with Bloch radius $r = 0.8$, and evaluates Fisher information locally at $\theta = 0$. For the illustrated equatorial family,

$$\rho_\theta = \frac{1}{2}[I + r(\cos \theta \sigma_x + \sin \theta \sigma_z)], \quad F_Q = r^2 = 0.64,$$
$$F_C(\varphi) = \frac{r^2 \sin^2 \varphi}{1 - r^2 \cos^2 \varphi}.$$

- The two vectors have a visible finite separation for illustration. The plotted information is **not** computed as their finite difference.
- At the initial basis, probabilities have zero first derivative at the reference point. They can still change at second order over a finite displacement.
- For $r < 1$, the CFI depends on the measurement basis as shown. In the pure-state limit $r \rightarrow 1$, generic projective bases in this plane attain the QFI; singular zero-probability settings require care with limits.
- The optimal basis can depend on the unknown parameter. Local attainability does not imply one globally optimal, immediately known measurement.

Reference: Appendix J of the deck and [measurement_overlay.tex](#).

Slide 5 — How we treat mixed states

Spoken script

What are mixed states? They describe statistical mixtures of quantum states, using a density matrix.

Why study them? Real sensors interact with their environment, and we may observe only part of the probe. Mixed-state sensitivity is therefore important, but computing it exactly becomes expensive as the system grows.

[Point to the truncated density matrix.]

Our approach keeps the dominant eigencomponents instead of treating the entire spectrum in equal detail.

VQSE, the Variational Quantum State Eigensolver, learns a change of basis that exposes the leading eigenvalues and allows their eigenvectors to be prepared.

These components then enter TQFI, Truncated Quantum Fisher Information, through fidelity-based bounds. Intuitively, we compare nearby states using a compact spectral description to estimate how distinguishable they are.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Fidelity is a similarity measure for states. We estimate how quickly nearby states become distinguishable, using selected spectral components and accounting for the part we did not retain. Truncation simplifies the description; it is not permission to ignore its uncertainty.

$$\rho_m = \sum_{i=1}^m \lambda_i |v_i\rangle\langle v_i|, \quad \lambda_1 \geq \dots \geq \lambda_d.$$

- A mixed state need not represent ignorance about a unique underlying ensemble: ensemble decompositions are nonunique. Entanglement with an unobserved system also produces a mixed reduced state.
- **Truncation is a chosen computational strategy, not a requirement for describing mixed states.** The matrix above is generally subnormalized: $\text{Tr } \rho_m \leq 1$. Renormalizing it would change the quantities entering the stated bounds.
- Large eigenvalue weight does not automatically mean that all parameter sensitivity is retained. Dependence of both eigenvalues and eigenvectors matters.
- The thesis treats truncated-fidelity and generalized-fidelity bounds at finite displacement; their local-QFI interpretation requires the appropriate small-displacement limit. TQFI is not simply “calculate ordinary QFI after deleting small eigenvalues.” Use Appendix A if challenged.
- VQSE changes the measurement basis, not the input state's eigenvalues. Its numerical outputs are approximate and depend on ansatz expressivity, optimization and sampling. The established method is from the literature; your contribution is its implementation, benchmarking and integration.
- The introduction calls mixed-state QFI less explored. Do not turn that into “nobody has studied mixed states.”

Sources: [Methods.tex](#), [VQSE.tex](#).

Slide 6 — Parallel numerical and quantum-circuit workflow

Spoken script

This is the computational structure I developed.

[Follow the upper row.]

Numerically, I prepare the state, restrict access and calculate its spectrum. This gives a reference for the circuit route below, where VQSE learns the dominant components needed for the information bounds.

The stages are modular, so I can check approximations against their numerical counterparts.

[Point to VQSE.]

I also tested a deterministic Bures natural-gradient update on a small mixed-state VQSE benchmark. The intuition is to scale parameter updates by how much they change the quantum state.

Fisher geometry therefore plays two roles: it quantifies distinguishability for sensing, and it can guide the optimization used in spectral estimation.

Details to keep in mind

Intuition — preparation only, not extra spoken text: A training parameter is a knob. Equal turns of two knobs need not move the quantum state equally far. Natural gradient accounts for that unequal effect when choosing an update; the sensing QFI and the training metric refer to different parameters.

- Partial trace obeys ($\rho_S = \text{Tr}_E \rho_{SE}$). Marginal measurement statistics on S agree whether E is traced out mathematically or simply unobserved. Conditioning on an outcome in E is a different operation.
- Interchangeable implementations do **not** imply interchangeable stage order; channels generally do not commute.
- Exact matrix calculations, Trotterized dynamics and variational spectral estimation have different approximation errors. “Validated against numerical references” is more defensible than an unconditional “perfectly matching.”
- Unitary evolution of a pure state does not prepare a global Gibbs state. The dynamical circuit example and the ground/Gibbs preparation branches are distinct.
- Do not claim a demonstrated hardware quantum advantage. This establishes and benchmarks a route to variational estimation.
- The updated VQSE benchmark compares BFGS, Adam and a deterministic damped Bures natural gradient on a small mixed-state problem. It supports a scoped numerical comparison, not a general speedup claim.
- Natural gradient uses a state-space metric, whereas Newton's method uses the objective Hessian. The relevant metric is with respect to trainable ansatz parameters, not the sensed magnetic field. Here the Bures metric treats the mixed output state directly; a scalable hardware version would need an appropriate sampled estimator. See Appendices B and C.1 and [VQSE.tex](#).

Source: [Methods.tex](#).

Slide 7 — Quantum Magnetometry

Spoken script

Let me first show what this framework tells us about a quantum magnetometer.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Use this divider as a breath and a change of subject. There is no extra concept to explain here.

This is a transition slide. Advance immediately after the sentence.

Slide 8 — A noisy quantum magnetometer

Spoken script

Quantum magnetometry has real applications: NMR probes monitored accelerator fields at CERN's LEP, and ESA's Juice carries a rubidium-based quantum magnetometer.

Here I use a controlled interacting-spin model, the transverse-field Ising chain. The ring represents spin couplings, not probes placed around an accelerator.

[Point to the six sites, then the two dashed sites.]

The field changes the probe state. Of six spins, only four are accessible; tracing out the other two can leave a mixed state.

I compare ground-state preparation, thermal preparation and depolarization. The aim is to understand which preparation gives useful sensitivity over the intended field range.

Details to keep in mind

Intuition — preparation only, not extra spoken text: A reduced state gives exactly the predictions needed for measurements on the spins you can access, even when those spins are correlated with the rest. It is not the state obtained by physically removing two spins and allowing the remaining system to re-equilibrate.

The plotted examples use ($N=6$), ($n=4$), periodic boundaries and ($J=1$). The general methods chapter also discusses open chains; do not mix their Hamiltonian summation limits with this ring.

For a fixed, parameter-independent partial trace or noise channel, QFI cannot increase. More accessible qubits can recover information hidden in the larger probe. However, changing the **preparation family** is not an application of that same data-processing comparison.

This is a finite model: say “sensitivity peak” or “near the critical region,” not a true finite-system thermodynamic singularity. Here “local QFI” may also refer to the accessible subsystem; distinguish that from locality in the estimated parameter when answering questions.

Real-instrument references: [CERN's LEP NMR probes](#) and [ESA's MAGSCA aboard Juice](#). These motivate magnetometry; neither is presented as a realization of the thermal Ising-chain model.

Source: [NoiseAnalysis.tex](#).

Slide 9 — Thermal preparation broadens the sensitive range

Spoken script

Here, white is the reduced ground-state reference and green is the reduced thermal preparation at inverse temperature β equal to two.

[Point first to the white maximum, then to the shaded region on the right.]

The thermal state does not win at the highest ground-state peak. Its advantage appears away from that optimum, where the green curve remains above the white curve.

Imagine that the field is not known well enough to place the sensor exactly at its best operating point. A narrow, high peak may be less useful than a response that remains informative across the relevant interval.

For a future space sensor with controlled thermal preparation, this could motivate choosing temperature to match the expected field range.

Details to keep in mind

Intuition — preparation only, not extra spoken text: A sharp peak is useful when you know where to operate. A wider useful response can matter when the field is uncertain. The plots illustrate that trade-off; they do not compute a universal optimum over unknown fields.

- ($\beta = 1/(k_B T)$): larger beta means lower temperature. Plot labels use (βJ); the numerical convention is ($J = 1$).
- The global Gibbs state ($e^{-\beta H(h_x)}/Z(h_x)$) is prepared **before** tracing out two spins. It is not generally the Gibbs state of the isolated four-spin Hamiltonian.
- The shaded region marks pointwise higher QFI in this example. You did not establish a globally optimal Bayesian or minimax preparation for an arbitrary prior over fields. "Useful when the field is uncertain" is a design interpretation of the curves, not a completed prior-weighted optimization.
- Thermal populations depend on the field; this can supply information in addition to changes of eigenvectors. This is not evidence that arbitrary heating or parameter-independent noise improves QFI.

Sources: [NoiseAnalysis.tex](#), [NoiseAnalysisAppendix.tex](#).

Slide 10 — Cooling strengthens the sensitivity peak

Spoken script

Now I lower the temperature, increasing beta to five.

[Point to the green peak.]

For this preparation and accessible subsystem, the thermal peak is also higher than the reduced ground-state peak.

The intuition is that low-energy excited states are not necessarily useless background. Their populations change with the magnetic field, and that change can carry information.

Together, these plots show two possible benefits: broader sensitivity away from the ground-state optimum, or stronger sensitivity around a favourable operating region.

This is not a universal claim that mixed states are better. It shows why preparation must be evaluated against the intended sensing task, rather than judged only by purity or by proximity to the ground state.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Thermal preparation gives several energy levels field-dependent populations. Those populations can act as additional indicators of the field. Their benefit must be compared with the other changes caused by temperature.

For the reported ($N=6, n=4, \beta=5$) periodic example, the thesis gives a 23.5% higher peak. Keep this as a question-answer detail; the main slide deliberately avoids large numerical callouts.

The comparison is between the **reduced ground-state family** and the **reduced globally thermal family**. Both accessible states can be mixed. The field derivative acts on the field-dependent Gibbs preparation, including its populations. No fixed noisy processing of the same ground-state family is claimed to increase QFI. The preparation resources and equilibration time are not optimized by these plots.

Slide 11 — Exoplanet Detection

Spoken script

The second application moves from a controlled sensor model to an optical experiment.

Details to keep in mind

Intuition — preparation only, not extra spoken text: This divider changes the physical platform while preserving the question about accessible information.

Change pace slightly; this is the start of the second application, not a new mathematical introduction.

Slide 12 — The exoplanet problem

Spoken script

A faint companion close to a bright star is difficult to resolve because their images overlap.

[Point from the unresolved image to the mode sorter.]

Direct imaging records photon positions. SPADE instead sorts the light into spatial modes.

With the sorter aligned to the star, most of the bright light enters the fundamental mode. The displaced companion also contributes to higher modes, where its weak signal is easier to detect.

This connects to the earlier measurement animation: changing the measurement basis can reveal information that another measurement misses.

Details to keep in mind

Intuition — preparation only, not extra spoken text: The optical sorter combines amplitudes before detection. That changes what is measured; merely processing the resulting camera intensities differently would not generally implement the same measurement.

The slide labels “Bright star” and “Exoplanet” are a visual shorthand for the dominant contributions. The physical outputs are spatial modes. The companion also contributes to the fundamental mode.

SPADE does not defeat all diffraction limits or identify the source of each photon. Its advantage is parameter- and model-dependent measurement sensitivity. Alignment, brightness ratio, throughput, background and readout matter.

Mutual incoherence of star and companion does not mean that their spatial density matrix is diagonal in an arbitrary HG basis. An individual displaced source occupies a coherent superposition of HG modes; an incoherent source mixture can retain off-diagonal entries in that basis.

Sources: [Astrophysical.tex](#), [ExoplanetExperiment.tex](#).

Slide 13 — SPADE modes and readout cross-talk

Spoken script

[Point to the fundamental and first-order modes.]

P-zero is the probability of the fundamental outcome. P-one combines the two first-order outcomes, which respond to the source separation.

The detector can confuse the reported ports. In this model, chi describes that cross-talk.

The reported first-order signal therefore includes some events from the bright fundamental port.

Even a small leak matters because the star is so much brighter. This gives us a simple classical description of the readout. Next, we need to ask what the available counts can tell us.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Suppose an illustrative acquisition sends 10,000 events to the fundamental outcome and 10 to the first-order outcome. A 0.1% leak from the bright outcome contributes about 10 false first-order events—already comparable to the weak signal. These are illustrative numbers, not the measured dataset.

For ideal mode outcomes,

$$p_0 = \text{Tr}(M_0\rho), \quad p_1 = \text{Tr}(M_1\rho),$$
$$M_0 = |HG_{00}\rangle\langle HG_{00}|, \quad M_1 = |HG_{01}\rangle\langle HG_{01}| + |HG_{10}\rangle\langle HG_{10}|.$$

The slide uses an assumed symmetric readout model. The pair $(\chi, 1 - \chi)$ gives the weights for reporting outcome 1 from inputs 0 and 1. It is one row of the full confusion matrix, not a quantum state or a complete channel by itself.

The symmetric readout assumption gives

$$p_1^{\text{rep}} = \chi p_0 + (1 - \chi)p_1.$$

- The companion also contributes to (HG_{00}) . Neither port exclusively identifies photon origin.
- Higher modes and loss require an additional outcome; generally $p_0 + p_1 \neq 1$ unless a binary normalization/conditioning convention has explicitly been imposed.
- Chi here is **classical reported-port confusion**, not coherent amplitude mixing. The latter requires a pre-measurement optical model.
- The thesis takes 0.0035 from an external reported cross-talk calibration. Equal reverse leakage and an unaffected failure outcome are assumptions, not a fully measured transfer matrix.
- A fixed classical post-processing channel can reduce CFI but does not change the pre-measurement state's QFI.

Slide 14 — What the experiment recorded

Spoken script

The ASI Matera experiment had already finished when we received these data.

[Point to the individual acquisition and the averaged map.]

At each setting, we have one hundred repetitions of the combined first-order count.

My initial aim was to use these measurements for TQFI. But the archive did not contain the separate modal outcomes or the measurements needed to recover the state's dominant eigencomponents.

The data did support fitting the average count response. The question became how to connect that fit to a physical description of the receiver.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Repeating the same measurement sharpens the mean without necessarily telling you more kinds of things. A hundred repeated totals do not automatically reveal the separate contributions to each total.

The record has 525 settings, 100 repetitions per setting and 10 ms per exposure: 52,500 observations, not necessarily 52,500 photons. Separate (HG_{00}) , (HG_{01}) , (HG_{10}) , higher-mode and loss records are absent.

Without an independent incident-flux calibration, count scale does not separate flux from efficiency. Without basis rotations or phase-sensitive observables, the record does not identify HG coherences. Repetition improves knowledge of the measured mean but does not add a new kind of observable.

The experimental TQFI protocol is not identifiable from this record. This does not prevent a conditional TQFI calculation from an assumed state and channel model. The fitted quantities describe the observed count response. They do not determine the complete optical channel.

Slide 15 — When modal counts become spectral information

Spoken script

This slide explains the gap between the original idea and the available data.

[Move from the measured populations to the middle panel.]

TQFI needs dominant eigenvalues and eigenvectors. SPADE gives populations in a fixed mode basis.

These would coincide if that basis matched the state's eigenbasis. In general, we cannot assume that it does.

So I kept the classical fit and developed a quantum description of the receiver. To explain that step, let us first connect probabilities to density matrices.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Looking at the diagonal of a matrix in one basis is not the same as diagonalizing the matrix. They agree only when that basis is an eigenbasis.

SPADE gives access to mode populations

$$p_j = \langle HG_j | \rho | HG_j \rangle.$$

If

$$\rho = \sum_j \lambda_j |HG_j\rangle \langle HG_j|,$$

then $p_j = \lambda_j$ and the retained approximation is

$$\rho_r = \sum_{j=0}^{r-1} p_j |HG_j\rangle \langle HG_j|, \quad \text{rank}(\rho_r) \leq r.$$

- An HG cutoff is a basis truncation. To reproduce the dominant rank- r spectral truncation used by TQFI, the chosen modes must be eigenvectors and must correspond to the r largest eigenvalues. Diagonality alone does not establish this ordering.
- Aggregating HG_{01} and HG_{10} into one value p_1 gives their sum, not two separate eigenvalues.
- “Normalized” on the slide means that the total trial normalization is known. It does not mean rescaling the retained probabilities to sum to one. The truncated operator may have trace below one. Keep the omitted weight when evaluating the

fidelity bounds. A normalized postselected state would answer a different question.

- Observing selected modes is a projection or coarse-graining operation; it is not the partial trace used in the magnetometer.
- VQ-SPADE proposes a trainable unitary that maps the state's eigenvectors onto resolved HG outputs. This is an ideal target that requires a sufficiently expressive transformation and successful optimization. It is kept for the conclusion and Appendix I.

Sources: [ExoplanetExperiment.tex](#), [Conclusions.tex](#).

Slide 16 — From probabilities to a quantum description

Spoken script

[Move from left to right.]

For two modes, we can collect their probabilities in a vector. We can also write those probabilities on the diagonal of a density matrix.

But an optical state can have additional entries: coherences between the modes. The same populations can therefore belong to different quantum states.

Classical cross-talk changes the reported probabilities. A quantum channel describes what happens to the optical state before measurement, including its coherences.

This gives us the language to model the receiver.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Modal probabilities describe how often outcomes occur. Coherences describe relations between modal amplitudes that a different measurement can reveal. Rewriting the probability vector as a diagonal matrix does not recover these relations.

The slide uses two **individual** modes, labelled a and b , with

$$q_a, q_b \geq 0 \quad q_a + q_b = 1 \quad |c|^2 \leq q_a q_b$$

The last condition ensures positivity of the two-mode density matrix. The symbols q_a, q_b are illustrative mode populations, not the scalar displacement quantity $q = d_a^2/4$ used in the source model.

- These are not the measured pair (p_0, p_1) from slide 13. There, p_1 combines two first-order modes and further modes or failure events may remain. The full receiver keeps a complement outcome.
- The middle and right panels compare possible state descriptions. They are not successive physical operations that create coherence from measured counts. Setting $c = 0$ is one possible state compatible with these populations.
- The probabilities belong on the diagonal of a **state**. The classical transition probabilities belong in the **channel matrix**. Do not embed the weights $(\chi, 1 - \chi)$ and describe the result as the optical state.
- The two-mode readout example is

$$R_\chi = \begin{pmatrix} 1 - \chi & \chi \\ \chi & 1 - \chi \end{pmatrix} \quad \mathbf{q}^{\text{rep}} = R_\chi \mathbf{q}$$

- In the actual low-mode model, the corresponding three-outcome matrix leaves the failure outcome unchanged. Both the symmetry and that isolation are assumptions.
- A stochastic matrix fixes only the classical transition probabilities. It does not determine how a quantum channel acts on coherences. Appendix F gives an explicit quantum extension and explains its non-uniqueness.
- A density matrix is a state. A quantum channel $\mathcal{E}$ is a map acting on density matrices. Merely writing a probability vector as a diagonal matrix does not specify such a map.

Source: [ExoplanetExperiment.tex](#), classical and quantum descriptions. Appendix F of the deck gives the technical extension.

Slide 17 — From two incoherent sources to an optical state

Spoken script

Where do those coherences come from in this experiment?

[Point to the two source contributions.]

The star and companion are independent sources, so we describe them as a statistical mixture.

However, the displaced companion's wavefunction extends across several HG modes. Those mode amplitudes can have definite phase relations, even though the two sources are mutually incoherent.

This source model therefore supplies both populations and modal coherences. We can now follow how the receiver acts on that state.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Each photon is attributed probabilistically to one of two mutually incoherent sources. Mutual incoherence removes interference between the source alternatives; it does not remove the spatial-mode structure of each source.

The conditional one-photon source model is

$$\rho_S(d_a, \epsilon) = (1 - \epsilon)|HG_{00}\rangle\langle HG_{00}| + \epsilon|\psi_{d_a}\rangle\langle\psi_{d_a}|.$$

For the one-axis Gaussian model,

$$|\psi_{d_a}\rangle = \sum_{n=0}^{\infty} c_n(d_a)|HG_n\rangle, \quad c_n = e^{-q/2} \frac{\alpha^n}{\sqrt{n!}}, \quad \alpha = \frac{d_a}{2}, \quad q = |\alpha|^2.$$

The ideal first-order population is consequently $p_1 = \epsilon q e^{-q}$ in the one-axis convention.

- There are no cross terms $|HG_{00}\rangle\langle\psi_{d_a}|$ between the two source alternatives.
- The term $|\psi_{d_a}\rangle\langle\psi_{d_a}|$ can nevertheless contain off-diagonal HG terms $|HG_m\rangle\langle HG_n|$ for $m \neq n$.
- The density matrix has rank at most two even though the displaced spatial wavefunction has support on infinitely many HG modes. Rank and basis support are different concepts.
- The numerical calculation uses a finite HG cutoff and a complement outcome. Cutoff convergence was checked independently.
- The slide is conditional on a Gaussian PSF, known programmed d_a and ϵ , mutual source incoherence and alignment to the bright source.

Sources: [Astrophysical.tex](#), [ExoplanetExperiment.tex](#).

Here $\epsilon = I_B/(I_A + I_B)$ is the faint-source photon fraction. The contrast I_B/I_A approximates it only when the companion is much fainter. The model uses $d_a = d/w_0$ for dimensionless separation.

Slide 18 — Why use quantum channels?

Spoken script

[Follow the sequence from left to right.]

We start from the optical state. Channels describe loss, alignment and changes to the modes before SPADE measures them.

For example, dephasing reduces coherence while leaving the mode populations unchanged.

SPADE produces outcome probabilities. Readout cross-talk, background and calibration then determine the predicted mean counts.

The benefit is that each physical effect has its own place. We can replace an assumption with a measured calibration and calculate how it changes the signal and the available information.

The present data test the count response. They do not identify every optical channel.

Details to keep in mind

Intuition — preparation only, not extra spoken text: A probability vector records the diagonal seen by one measurement. A quantum channel is needed when the device can also change the off-diagonal information that another measurement could reveal.

In a two-mode illustration,

$$\rho = \begin{pmatrix} q_a & c \\ c^* & q_b \end{pmatrix}, \quad \mathcal{D}_v(\rho) = \begin{pmatrix} q_a & vc \\ vc^* & q_b \end{pmatrix}.$$

The dephasing example uses a real visibility $0 \leq v \leq 1$. A direct HG measurement sees the same populations before and after this map. A later coherent mode transformation can turn coherence into a population difference. With a known transformation and suitable inputs, populations can therefore carry indirect information about dephasing. The present aggregate record and incomplete calibration do not identify v independently.

Classical readout cross-talk instead acts on the measurement output:

$$\mathbf{p}^{\text{rep}} = R_{\chi}\mathbf{p}, \quad p_1^{\text{rep}} = \chi p_0 + (1 - \chi)p_1.$$

Other stages remain part of the complete receiver model:

- Before measurement: common loss, alignment or jitter, dephasing, coherent mode mixing, mode-dependent loss and accessible-mode reduction.
- Measurement: $p_y = \text{Tr}(M_y \rho)$.
- After measurement: readout confusion, additive background, calibration variation and repeated-count statistics.
- A background parameter may contain dark counts, ambient background or unresolved leakage; the current record cannot separate them.
- The quantum-channel description is useful for propagation, QFI analysis and testing alternative measurements. It is not evidence that every channel parameter was learned from this dataset.

The Kraus formalism and its non-uniqueness are available in Appendices E, K and L if the quantum-information examiner asks. They are not needed in the main spoken explanation.

Source: [ExoplanetExperiment.tex](#), [ExoplanetExperimentAppendix.tex](#).

Common survival is counted once. Conditional modal probabilities are multiplied by $T = N_{\text{in}}\eta$. The example dephasing and mixing parameters are fixed sensitivity assumptions. They were not fitted from these counts. QFI comparisons use a common per-surviving-photon normalization and parameter-independent channels.

Slide 19 — A conditional model captures the observed mean pattern

Spoken script

[Compare the observed and modelled maps.]

With the optical assumptions fixed, we fit the observable count response.

The model captures the central minimum and the increase with separation and companion intensity.

A separate check tested the classical calibration model on settings left out of fitting. It predicted the means well. The detailed results are in the appendix.

This supports the model's practical use. The channel description also lets us study how optical imperfections affect sensitivity and identify which extra measurements would help.

It is foundation work for a future experiment with resolved modes and a controllable measurement basis.

Details to keep in mind

Intuition — preparation only, not extra spoken text: Predicting the average and explaining every fluctuation are different tasks. Two distributions can share a mean while having different spreads. Matching the mean is useful, but supports a narrower conclusion than identifying the full apparatus.

- The displayed panels are the **provisional workbench's observed and modelled setting means**, compared on the data used for fitting. Do not call this plot alone a held-out validation.
- The separate hierarchical calibration model was checked by holding out checkerboard cells, complete distance rows and complete source-ratio columns. Nominal 95% intervals covered 93.52%, 94.67% and 93.71% of the held-out setting means. These are the Appendix H results, not a validation statistic for the two provisional-workbench panels shown here.
- Reproducing means is not reproducing the distribution of individual counts. The thesis finds overdispersion and rejects the tested simple deterministic count models as complete descriptions of individual exposures.
- Avoid "the channels were fitted" and "I reconstructed the experimental QFI." Instead: **"The optical channels were specified; the observable count response was calibrated."** A model-based QFI remains conditional on the specified state and channels.
- More measurements help only if they resolve relevant ambiguities: separate ports address modal aggregation; known flux and loss outcomes address efficiency; phase-sensitive or rotated-basis measurements address coherence; controlled mode injection constrains transfer properties. Complete channel identification can require several calibrated inputs as well as outputs.

The prediction target is the mean of 100 repetitions, expressed as counts per 10 ms. The Gaussian-process correction is classical calibration variation. Mean prediction does not establish the distribution of every individual count or reconstruct experimental QFI.

Slide 20 — Conclusions

Spoken script

To conclude, I developed numerical and quantum-circuit methods for studying sensor sensitivity and estimating dominant state components.

For the magnetometer, the useful preparation depends on the field range where the sensor will operate.

For SPADE, the initial TQFI goal led to a calibrated count model and a physical channel framework. VQ-SPADE is the proposed next step, using a trainable optical transformation to learn the state's eigenbasis.

Across both applications, the central question is how preparation, access and measurement determine the information we can use.

Thank you.

Details to keep in mind

Intuition — preparation only, not extra spoken text: End by separating implemented methods, numerical findings, data interpretation and a future proposal. The common lesson is the chain from preparation to the recorded measurement.

- The numerical and circuit implementations were compared against exact references; finite numerical precision, Trotter error and variational convergence still matter.
- The slide's phrase “would diagonalize” describes the ideal goal of VQ-SPADE. A realizable device would learn an approximate eigenbasis, subject to optical control, calibration and optimization limits. The archived experiment did not implement it. Diagonalization alone is also insufficient for TQFI. Nearby states and additional overlap measurements are required.
- The main talk ends here. The following optional responses are outside the 15-minute budget.

Backup Slide — Appendix N: Quantum Cramér–Rao bound

Spoken script

For a fixed measurement, the classical Cramér–Rao bound says that the variance of a locally unbiased estimator is at least one over the number of independent repetitions times the classical Fisher information.

The QFI is the maximum classical Fisher information over all measurements. Replacing the measurement-dependent information by this maximum gives the quantum Cramér–Rao bound:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{N\mathcal{F}_Q(\theta)}, \quad \Delta\theta \geq \frac{1}{\sqrt{N\mathcal{F}_Q(\theta)}}.$$

It is therefore an ideal local precision benchmark. Reaching it still requires an appropriate measurement and an efficient estimator; the optimal measurement can itself depend on the unknown parameter.

Details to keep in mind

- The bound assumes a differentiable one-parameter model, independent repetitions, regularity and local unbiasedness.
- Since $\mathcal{F}_C(M) \leq \mathcal{F}_Q$, a particular measurement obeys

$$\text{Var}(\hat{\theta}) \geq \frac{1}{N\mathcal{F}_C(M)} \geq \frac{1}{N\mathcal{F}_Q}.$$

- For a single parameter, measuring in an SLD eigenbasis can attain the QFI locally under the usual conditions. This does not imply one fixed, globally optimal measurement for every parameter value.
- The QCRB concerns estimator variance, not absolute error in a single experimental run.

Backup Slide — Appendix C: Adaptive VQSE training

Spoken script

This plot makes the adaptive training schedule explicit. In the upper panel, the white curve is the objective optimized at each outer iteration; the dashed curves evaluate the same variational state against the local and global Hamiltonians separately.

The lower panel shows why the optimized objective changes during the run. We start with the local objective, then quadratically transfer weight to the global objective. The white vertical lines mark refreshes of the global target basis states.

Therefore the white curve is useful for monitoring the scheduled optimization, but it is not by itself a fixed-loss convergence curve. To assess spectral accuracy, we compare the estimated eigenvalues with the exact numerical spectrum.

Intuition

Think of the optimization as learning in two stages. The local Hamiltonian first gives the circuit an easier directional signal; the global Hamiltonian then evaluates whether the learned basis places the dominant spectral weight in the desired computational-basis states. The lower panel is the mixing dial between those two tasks.

- The state is a balanced, non-diagonal two-qubit mixed state with a rank-3 target. The result is a deterministic simulator demonstration, not a hardware benchmark.
- A global-target refresh does not necessarily create a visible jump: a jump occurs only if the selected target basis states actually change.
- The adaptive cost changes because the Hamiltonian changes. Use the leading-eigenvalue error on the next slide for a fixed accuracy metric.

Backup Slide — Appendix C.1: BFGS, Adam and Bures natural gradient

Spoken script

Here I compare three optimizers from the same initial circuit parameters on the same adaptive VQSE problem: BFGS, Adam, and a deterministic damped Bures natural gradient.

[Point to the upper-left panel.]

This is the fixed metric: the L2 error of the three leading estimated eigenvalues relative to the exact spectrum. BFGS and the Bures natural gradient both reduce it to the few- 10^{-3} range in this run, while Adam remains farther away.

[Point to the lower-left panel.]

The final-spectrum panel shows the same outcome directly: BFGS and Bures-NG closely reproduce the three exact leading eigenvalues.

[Point to the lower-right panel.]

The most visible difference is in parameter motion. BFGS has intermittent large and small Euclidean updates, whereas the Bures natural-gradient updates become smooth and steadily decrease after the initial adjustment.

The reason is geometric preconditioning. The update is

$$\Delta\theta = -\eta(G_{\text{Bures}} + \lambda I)^{-1}\nabla C.$$

The Bures metric is the mixed-state geometry associated with quantum Fisher information. It rescales directions by how much they move the physical density matrix, rather than treating every circuit parameter as having the same physical scale.

Intuition

Two circuit knobs can be turned by the same angle while moving the quantum state by very different amounts. Ordinary parameter-space updates can therefore overstep in sensitive directions or make little progress in redundant ones. The Bures metric acts like a map of those unequal distances: damping prevents unstable inversion, and the inverse metric rescales the gradient before the update.

- The upper-right adaptive objective should not be read as a fixed-loss race, because the local-to-global Hamiltonian schedule changes it over time.
- BFGS is also curvature-aware; its jagged update norms here are not proof that BFGS is intrinsically unstable. In this benchmark it is restarted at each outer adaptive stage while the objective changes.
- The smoother Bures-NG trajectory is an observation for this state, ansatz and damping choice, not a general guarantee of faster convergence.
- This exact Bures-metric construction uses full density matrices and is appropriate for a small deterministic simulator. A hardware version would require a sampled metric estimator or another scalable approximation.

Optional responses for questions

"Is the quantum natural gradient just Newton's method?" — Appendix B

"They share the idea of transforming the gradient using a matrix. But Newton's method uses curvature of the objective, while the quantum natural gradient uses the geometry of the state family. It scales a parameter step by how much that step changes the quantum state."

$$\Delta\alpha = -\eta(F^{\text{ans}} + \lambda I)^{-1}\nabla C.$$

Here (F^{ans}) is a metric with respect to **trainable circuit parameters**, not the scalar QFI with respect to the sensed field. For pure states, the QFI matrix is four times the Fubini–Study metric, depending on convention. A mixed-state VQSE input needs an appropriate mixed-state metric; a pure-state metric is otherwise a surrogate. Regularization stabilizes poorly conditioned directions. Do not claim a measured general speedup or equivalence to the Hessian.

“Why can temperature help if noise cannot increase QFI?” — Appendix D

“The comparison changes the preparation family. At each field, the Gibbs populations depend on that field, so they can carry sensitivity. This is different from applying the same parameter-independent noisy channel to an already prepared ground-state family.”

“Does retaining most probability retain most information?” — Appendix A

“Not automatically. A small component can vary strongly with the parameter. The retained spectral weight gives a compact representation, but metrological accuracy has to be assessed using the bounds and the numerical benchmark.”

“Did you identify the detector’s quantum channel?” — Appendices F, H, K–M

“No. The counts constrain an observable classical response. I constructed compatible, physically motivated optical stages and calibrated the count model, while keeping their assumptions explicit. Identifying the optical channel would require additional controlled measurements and calibrations.”

“Why not just use a classical regression?”

“A classical model is appropriate for predicting the measured counts, and the thesis includes that layer. The channel framework adds a description of the optical stages before measurement. It lets us ask how a different measurement, loss mechanism or calibrated optical transformation would change the accessible information. The current data do not uniquely determine that optical description.”

“What does VQ-SPADE add to ordinary SPADE?” — Appendix I

“Ordinary SPADE measures in a specified modal basis. VQ-SPADE would add a trainable optical transformation so that the measurement basis could be adapted towards an eigenbasis of the state. That would need resolved outputs and calibration, and remains a proposed extension.”

Rehearsal notes

- First rehearse **only** the spoken sections, including the five overlay advances on slide 3. The technical notes and intuition boxes are not a second spoken paragraph.
- Clock checkpoints: start magnetometry near **5:05**, exoplanets near **7:55**, and conclusions near **14:10**.
- If running late, omit the modularity sentence on slide 6 and shorten the future-measurement examples on slide 18. Preserve the distinctions between populations/eigenvalues, assumptions/fits, and demonstrated/proposed work.
- On plots, name the axes or curves, point to the relevant feature, then state its implication. Give the audience a moment to look before continuing.
- Do not rush the final sentence. Finish with the conclusion slide visible.

Further question preparation

See [defence_questions.md](#) for 60 questions with short oral answers, intuitive explanations and technical follow-ups. Especially useful follow-ups are questions 23–28: eigenbasis versus sensing basis, unknown parameters, thermal preparation, coherence, fidelity and photon-resource accounting.

For a slower introduction to thermal preparation, see questions 33–52 in [defence_questions.md](#), including a two-level example and a 30-second oral explanation. These are preparation notes, not additions to the timed talk.

Real-instrument examples and suggested slide wording are in questions 53–60 of [defence_questions.md](#). They distinguish MAGSCA and NMR probes from the periodic Ising model, and explain the preparation requirements for a possible temperature-controlled space sensor.